

AI Stock Trading Platform

UG_HF_03

Andrew Butler, Matthew He, Ethan Roe, Aaron Trelstad

Project Overview

- Customer Requirements

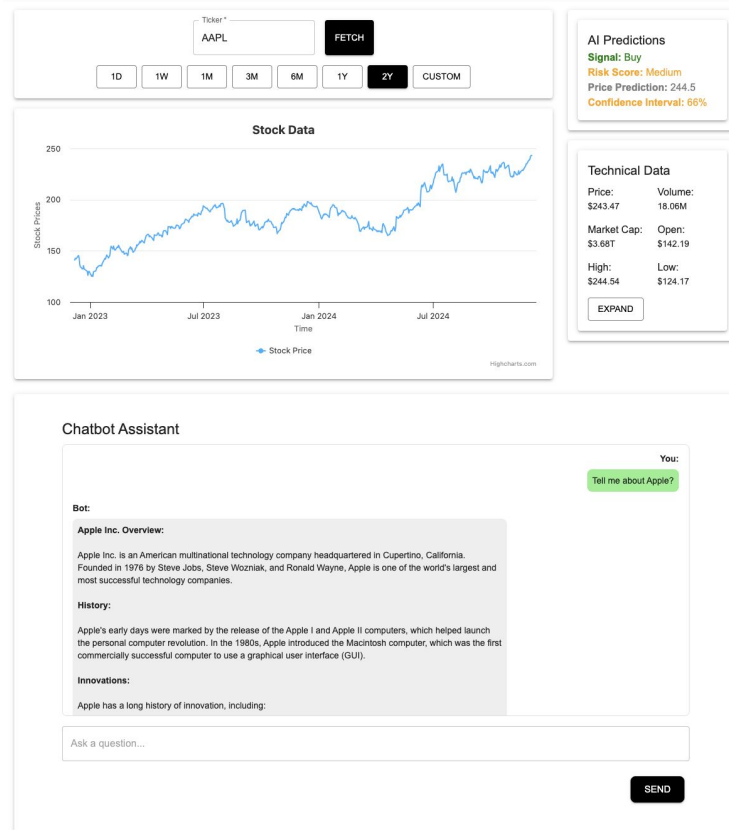
- Market Simulation
- Trade Simulation
- Predictive AI Features
- Student Learning Tool

- Feature Roadmap

- Emulates Investing App (Fidelity, Robinhood)
- Stock Tracking Dashboard (Apple Stocks App, Yahoo Finance)
- AI Features
 - Market trend detection
 - Support and Resistance Levels
 - Risk and Reward Calculations

Features

- Inspired by BunnyQuant
- User Stock Trading Dashboard
 - Visualize stock data
 - Show trade information
- AI Support for:
 - Market trend detection
 - Buy/Sell Recommendations
 - Support and Resistance Levels



Features

What You Get

Technical Analysis

- Trend Detection & Direction
- Support & Resistance Levels
- Pivot Points (Standard, Fibonacci, Camarilla)
- Advanced Cycle Detection
- Distribution Analysis
- Risk/Reward Calculations

Fundamental Analysis

- Discounted Cash Flow (DCF) Valuation
- Financial Ratios & Metrics
- Revenue & Earnings Growth
- Balance Sheet Analysis
- Intrinsic Value Calculation
- AI-Powered Insights

Price Alerts

- Price Above/Below Triggers
- Support/Resistance Breakouts
- Trend Reversal Alerts
- Email Notifications
- Multi-Asset Monitoring
- Real-Time Updates

Advanced Features

- Auto-Refresh Data (Premium+)
- Fractal Pattern Recognition
- Adaptive Signal Detection
- Multi-Timeframe Analysis
- Custom Analysis Workflows
- Portfolio Analytics (Enterprise)

Overall Trend | Market Cycle | Adaptive Signals | Risk & Position Intelligence | Multiframe Analysis | Seasonal Patterns | Pivot & S/F

Overall Trend

Downtrend (momentum weakening)

Technical Signal Summary

- MACD: Downtrend momentum
- RSI: RSI falling (bearish weakness)
- Bollinger: Price breaking below lower band (oversold / breakdown)
- A/D: Selling pressure (A/D confirms downtrend)
- StochRSI: Oversold

How to Interpret

MACD: Momentum indicator. Bullish crossover = buy signal, bearish crossover = sell signal.

RSI: Overbought (>70) may indicate correction, oversold (<30) may indicate rebound.

Bollinger Bands: Price breaking bands suggests volatility expansion.

A/D Line: Volume confirmation. Divergence = potential trend reversal.

StochRSI: Sensitive momentum indicator for short-term trades.

Technologies

- Current

- Frontend
 - React.js
 - Lightweight Charts
- Backend & APIs
 - FastAPI
 - Finnhub Stock API

- Planned

- LLM Serving
 - vLLM
- Database
 - MariaDB

Sample Finnhub API data

```
{
  "country": "US",
  "currency": "USD",
  "exchange": "NASDAQ/NMS (GLOBAL MARKET)",
  "ipo": "1980-12-12",
  "marketCapitalization": 1415993,
  "name": "Apple Inc",
  "phone": "14089961010",
  "shareOutstanding": 4375.47998046875,
  "ticker": "AAPL",
  "weburl": "https://www.apple.com/",
  "logo": "https://static.finnhub.io/logo/87cb30d8-80df-11ea-8951-00000000092a.png",
  "finnhubIndustry": "Technology"
}
```

```
1  {
2    "data": [
3      {
4        "p": 7296.89,
5        "s": "BINANCE:BTCUSDT",
6        "t": 1575526691134,
7        "v": 0.011467
8      }
9    ],
10   "type": "trade"
11 }
```

Challenges

- LLM Interaction
 - LLM needs to output data in correct format
 - LLM team collaboration to get what is needed for frontend
- Data acquisition
 - Limited API calls
- Client Communication
 - Client is a graduate student
 - Busy, so longer response times